

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2 – Manipulation (BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO5 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)		
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Section / Batch:	S/bt-D1 2024	Date:	4/8/26.

Code of Conduct

I K. Manoj Kumar (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Frequency Domain Image Enhancement	Periodic noise removal, Fourier Transform, notch filtering, quality inspection	1	10
Spatial Filtering & Edge-Preserving Enhancement	Bilateral filtering, edge preservation, autonomous vehicle navigation	2	10
GRAND TOTAL:			20 Marks

Annexure A – Scenario Based Assessment

Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

- (a) Interpret the scenario and identify the type of noise present.
- (b) Explain how this degradation affects inspection accuracy.
- (c) Recommend an appropriate enhancement technique to remove the interference.
- (d) Justify why your selected approach is more suitable than spatial-domain methods.
- (e) Explain your answer in a logical sequence.

Scenario 2: Autonomous Vehicle Navigation

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

- (a) Interpret the scenario and explain the challenge involved.
- (b) Identify the importance of preserving edges while removing noise.
- (c) Recommend a suitable filtering technique.
- (d) Justify why your selected filter preserves important image details.
- (e) Present your explanation clearly and systematically.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
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Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Scenario 1: Manufacturing Quality Inspection:

a) Interpret the scenario and identify the type of noise present.

The automated inspection system captures images of metal components, but repetitive horizontal stripe patterns appear due to electrical interference.

This type of degradation is periodic noise.

which is usually introduced by electrical or electromagnetic interference during image acquisition. periodic noise appears as repeated horizontal or vertical stripes and mainly affects.

b) Explain the this degradation affects inspection accuracy

Periodic noise reduces the visibility of small defects such as scratches, cracks, dents, or surface imperfections.

- * false detection of defects.
- * missing actual defects.
- * Reduced accuracy and reliability of quality inspection.

c) Recommend an appropriate enhancement technique to remove the interference.

Frequency Domain method.

* Fourier Transform to convert the image into the frequency domain.

* notch filter to remove the frequency components responsible for periodic noise.

d) Justify why your selected approach is more suitable than spatial-domain methods.

The frequency-domain approach is more effective because periodic noise is concentrated at specific frequency locations. A notch filter can selectively remove only these unwanted frequencies while preserving useful image details.

Spatial-domain filters such as mean or median filters process all pixels equally and cannot selectively remove periodic interference.

Explain your answer in a logical sequence.

- * Image is captured from the inspection system.
- * electrical interference introduces periodic horizontal stripes.
- * Apply the Fourier Transform to convert the image into the frequency domain.
- * Apply a notch filter to remove these unwanted frequencies.

Conclusion: frequency-domain enhancement with a notch filter is the best solution. it removes periodic quality effectively while maintaining image quality reliable defect detection.

2) Autonomous Vehicle Navigation:

a) Interpret the scenario and Explain the challenge involved.

The Self-driving car captures road images during heavy rain. Rain introduces random noise into the images, while important features such as lane markings and road boundaries must remain clearly visible. The challenge is to remove the noise without blurring.

b) Identify the importance of preserving Edges while removing noise:

Edges contain important structural information about the road environment.

- * Detect lane marking Accurately.
- * Identify road boundaries
- * Recognize obstacles.
- * make safe steering and navigation decisions.

c) Recommend a suitable filtering technique:

The bilateral filter smooths noisy regions while preserving strong edges by considering both.

- * spatial distance b/w pixels.

- * Intensity difference b/w neighbouring pixels.

d) Justify why your selected filter preserves important image details.

Unlike Gaussian or mean filters, the bilateral filter avoids averaging across edges. It smooths only pixels with similar intensity values, thereby reducing noise while keeping lane markings and road edges sharp.

e) Present your explanation clearly and systematically

- * The vehicle captures a road image during rainfall.

- * Rain introduces random image noise.

- * Apply a Bilateral filter to reduce noise.

- * The filter smooths homogeneous regions while preserving edges.

Conclusion: The bilateral filter provides the best balance b/w noise reduction and edge preservation making it ideal for Autonomous vehicle navigation in rainy conditions.